Duarte Fernández Piñeiro

ARTIFICIAL INTELLIGENCE ENGINEER

University of Vigo graduate · Class of 2026
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LinkedIn · GitHub · Portfolio with AI assistant
English B2 · French B1



What if my CV could answer your questions? My AI assistant can. Try it on my portfolio.

PROFESSIONAL PROFILE

A graduate of the University of Vigo's first cohort in Artificial Intelligence. I enjoy tackling problems without an obvious solution: understanding them thoroughly, exploring alternatives and building practical solutions. I have developed end-to-end projects in generative AI, RAG and automation. I am resourceful, independent, curious and persistent; I stay calm when facing difficulties, work well in a team and enjoy learning.

PROFESSIONAL EXPERIENCE

Artificial Intelligence Engineer | AHORA Enterprise · *Feb. - Jul. 2026*

- **Hybrid RAG for corporate documentation.** I developed a RAG system combining lexical (BM25) and semantic (embeddings) search, with reranking to improve information retrieval. Integrated with enterprise software and designed with a scalable, multi-user architecture ready for local deployment. View project
- **AI solution evaluation.** Research and analysis of AI alternatives to improve efficiency and reduce company costs.
- **Sales analysis and forecasting.** Development of an intelligent system to process business data and forecast future sales, aiming to replace reporting and processes previously handled in Power BI.
- **Computer vision.** Development of an AI model for a client to analyse photographs of building facades and identify the products needed for work on them.

SELECTED PROJECTS

- **Computer vision cooking algorithm** · View project
  Development of an architecture for universal automated cooking. It visually analyses a basket, identifies individual pieces and recommends cooking times and temperatures. It combines deep learning, computational geometry, probability calibration and statistical evaluation with data preparation and experimental evaluation. Its design supports adaptation to different foods and appliances.
- **Personal portfolio with AI assistant** · View project
  Development of a website with a professional assistant to answer questions about my profile. I used the project to explore AI-assisted development and apply concepts including Harness Engineering, Agent Skills, MCP, Spec-driven Development, SDD taxonomy, and the coordination of agents and subagents.
- **Implementation of ConvNeXt from scratch** · View project
  Recreating the ConvNeXt architecture from its original research paper, without using AI assistance or prebuilt model components. This project explored the mathematical foundations of convolutional neural networks, their internal workings and the mechanisms involved in learning.
  - **Sentiment classifier for Bitcoin-related messages** · View project
    Design of a web application with FastAPI that analyses Bitcoin-related messages using Sentence Transformers and LightGBM, generating contextualised recommendations with OpenAI. It achieved 87.5% accuracy and an F1 score of 88.1% on the test set, with deployment prepared using Docker and Cloud Run.

TECHNICAL SKILLS

GENERATIVE AI AND RAG

LLMs, RAG, LangChain, LangGraph, MCP, OpenAI API, embeddings, BM25, RRF, ChromaDB, Sentence Transformers, Jina Reranker and Ollama.

MACHINE LEARNING AND VISION

PyTorch, scikit-learn, LightGBM, OpenCV, ConvNeXt, DINOv3, YOLO, SAM, SegFormer, transfer learning, fine-tuning, model evaluation and calibration.

SOFTWARE AND DATA

Python, SQL, PostgreSQL, MongoDB, TypeScript, Java, FastAPI, REST APIs, Pydantic, pgvector, pandas, NumPy, Git, automated testing and Power BI.

INFRASTRUCTURE AND MLOPS

Docker, Linux, DigitalOcean, MLflow, ONNX, vLLM, model deployment and optimisation.

EDUCATION

Bachelor's Degree in Artificial Intelligence · University of Vigo, 2026 · first cohort | AI-Assisted Development Course | Microsoft Applied Skills: Develop a Generative AI Chat App Using the Microsoft Foundry SDK